SEMESTER 2/20252026

SPECIAL TOPIC IN DATA ENGINEERING (SECP3843)

SECTION 01

TUTORIAL MS AZURE

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Project Overview

This project was initiated to fill a business gap in the lack of understanding of the demographics of the customers, namely how gender is distributed among customers and how it might affect the product purchasing behavior. Customer and sales data were stored in an on-premise SQL Server database but were not being leveraged to do analytical work with. The important stakeholders demanded a detailed KPI dashboard that would help to understand the sales by gender and product category including total number of products sold, total number of sales revenue and the proportion of gender in the customer base. Moreover, the dashboard had to be able to filter by product category and customer title to allow more specific trends analysis. This solution is particularly relevant to retail and e-commerce businesses where customer demographic analysis directly informs marketing strategies and product positioning.

To meet these needs, an end-to-end data engineering pipeline was deployed in the Microsoft Azure cloud platform. The main goal was to copy data out of the local SQL Server database, load it into the cloud storage, convert it into a clean, aggregated format, and ultimately deliver it in the form of an interactive Power BI dashboard. The pipeline was also to run automatically on a daily basis to make sure the stakeholders were always in a position to access the latest information. This project revealed how several Azure services can be combined to form an entire and working data solution.

Description on the Solution

The process started by the provisioning of the needed resources in the Azure ecosystem, with the help of the available Azure for Students subscription. Since this type of subscription has restrictions on its deployment in the regions, the Japan East region was chosen as the location of deployment after it was tested that it was compatible with the services it needed. A special resource team, called intech-rg was formed to handle all the associated resources.

The initial stage of the pipeline was concerned with data ingestion. Azure Data Factory was set up to transfer data in the local AdventureWorksLT2022 SQL Server database into Azure Data Lake Storage Gen2. The raw data was deposited in a container called the bronze layer that maintained the data in the exact format it was in the source system without any alterations. This approach ensures that a complete, unaltered copy of the source data is always available for reprocessing if needed.

After the ingestion, the data was transformed using Azure Databricks. PySpark notebooks were created to read the raw data in the bronze layer, carry out cleaning processes like how to deal with missing values and how to standardize the column names and write the cleansed data in the silver layer. Additional aggregation and transformation logic was used to create business-ready tables, like sales summaries by product category and customer title, which were stored in the gold layer.

The last serving phase involved the creation of external views on the Delta tables stored in the gold container with the help of the Azure Synapse serverless SQL. Synapse serverless SQL was chosen because it allows querying data directly from Data Lake Storage without requiring a dedicated SQL pool, which reduces costs for this project. To enable generation of dynamic views, a stored procedure, CreateSQLServerlessView_gold, was designed. These Synapse views were then linked to Power BI and used to create an interactive dashboard. The dashboard visualizes the most important measures such as total products sold, total sales revenue, and gender distribution and offers slicers to sort by product category and customer title. A complete pipeline was to be executed on a daily basis to make sure that the dashboard shows the most recent available data.

Technology Used

This project used a mix of on-premise software, and Microsoft Azure cloud service to create an automated, end-to-end data pipeline. All technologies were chosen based on a particular stage of the data lifecycle, in order to extract data and visualize it:

SQL Server (On-Premises): SQL Server was the original source of data, where the raw AdventureWorks data were stored prior to being migrated to the cloud.

Azure Data Factory (ADF): Played the role of the main orchestration and ingestion engine. ADF used a Self-Hosted Integration Runtime (SHIR) to securely access the on-premises SQL Server to dynamically extract tables. It was also used to automate the entire pipeline process, where downstream Databricks notebooks were automatically triggered when the ingestion has been finished.

Azure Data Lake storage Gen2 (ADLS Gen2): It offers a scalable and optimized storage of big data analytics. It allowed the Medallion Architecture by hierarchical namespace, which subdivided data into three separate containers: bronze (which contains raw data in the form of Parquet files), silver (which contains clean data in the form of Delta tables), and gold (which contains fully transformed and analytics-ready Delta tables).

Azure Databricks: Worked as the distributed data processing engine. With the help of PySpark notebooks, Databricks standardized and formatted the data of the Bronze layer (Silver layer) to facilitate the business-level transformations, including the conversion of column names to snake case, and aggregated the final data (Gold layer).

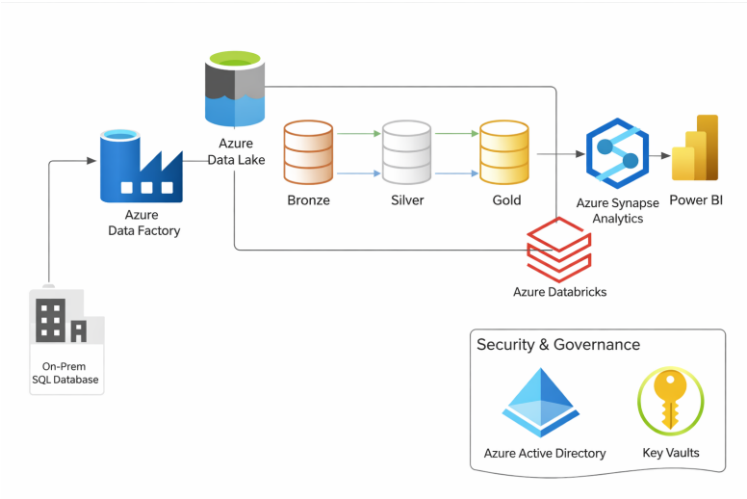
Azure Synapse Analytics: Provided as the data serving layer. Instead of storing data in a separate data warehouse, a Serverless SQL database was developed in Synapse. This enabled the generation of logical SQL views directly on the gold-layer Delta tables through OPENROWSET function, which saved processing time and storage costs.

Azure Key Vault: Provided pipeline security and best practice through serving as a safe depository of sensitive information. It securely stored the on-premises database credentials, the Databricks access tokens, and the Azure Data Factory could authenticate and check with these services without having to hard-code any secrets.

Microsoft Power BI: Served as the end-data visualization. It was connected to the Azure Synapse Serverless SQL endpoint to ingest the gold-layer views to enable manual data modeling (defining table relationships) and creating an interactive dashboard.

Data Pipeline Architecture

The data pipeline follows a layered medallion architecture, where data is refined from raw input into business-ready outputs. The diagram below shows the full end-to-end flow across all components.

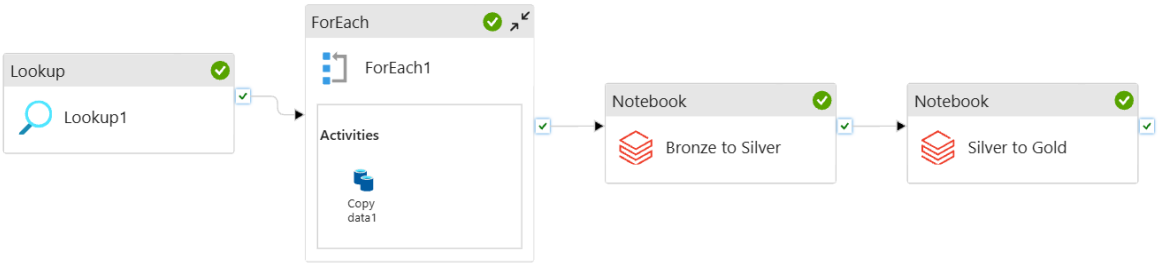


Stage 1: Data Source (On-Premises SQL Database)

The pipeline starts with an on-premises SQL Server database that stores customer and sales transactions. This acts as the main operational system, where daily business data is generated and maintained before being moved to the cloud.

Stage 2: Ingestion (Azure Data Factory)

Azure Data Factory handles data ingestion and overall pipeline orchestration. It connects to the on-premises database and extracts tables dynamically using a ForEach activity, allowing new tables to be included without modifying the pipeline. It also coordinates downstream processes, such as triggering Databricks notebooks, updating Synapse views, and scheduling pipeline runs. The diagram below shows the data pipeline is created in Azure Data Factory.



Stage 3: Storage & Transformation

The ingested data is stored in Azure Data Lake Storage Gen 2, organized into three layers:

- **Bronze Layer**: Stores raw data exactly as it is extracted, serving as a backup of original records.
- **Silver Layer**: Data is cleaned and transformed using Databricks (PySpark), including tasks like standardizing formats, renaming columns, fixing data types and removing duplicates.
- **Gold Layer**: Data is further processed into aggregated, business-ready tables optimized for reporting and analysis.

Stage 4: Serving Layer (Azure Synapse Analytics)

Azure Synapse Analytics in the Gold layer and provides SQL views for the processed data. This makes the data easier to query and removes the need to interact directly with raw files in the data lake.

Step 5: Visualization (Power BI)

Power BI connects to the Synapse views to create an interactive dashboard, allowing users to easily understand business insights and support better decision-making.

Stage 6: Security & Governance (Azure Active Directory + Azure Key Vault)

Security and access control are handled across the entire system using Azure Active Directory (Entra ID), which manages authentication between services. Azure Key Vault is used to securely store sensitive information such as connection strings and access keys, ensuring credentials are never hardcoded and are safely retrieved when needed.

MS Power BI Dashboard

This dashboard provides a high-level summary of business, highlighting 708.69K in Total Sales across a catalogue of 298 unique products. The data features a heavy demographic lean toward customers addressed as "Mr." (58.33%) and "Ms." (40.48%), as shown by the central doughnut chart, while the interactive slicers on the right allow users to filter these metrics by specific customer titles or product categories.

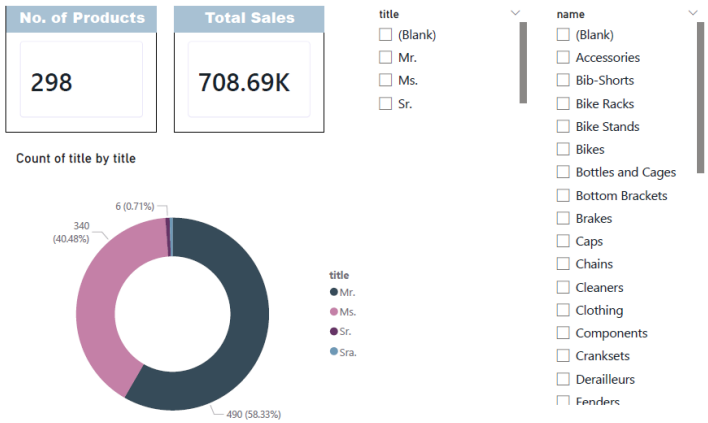


Figure shown dashboard of this project

Reflection

Lim Yu Han

During the implementation of this project, there were a number of technical issues that were faced and the process of troubleshooting taught me a lot of lessons in cloud data engineering. One of the major issues was the dbutils.fs.mount() command in Azure Databricks. The mount command kept failing because of the security configuration of the cluster. A solution to this was to create a new cluster with Single User access mode. I also got acquainted with the fact that using tutorials, software versions and updates to the platform can lead to differences that need to be resolved through independent troubleshooting. In addition to addressing certain technical issues, I was able to understand how various Azure services can be used together. The most important lesson during this project was to understand the interrelation and functionality of these services. Lastly, I was able to understand that data engineering in the real world is not only about following instructions. It involves having problem-solving skills, reading error messages carefully and having the desire to find solutions when things fail to work according to plan. They are the abilities that cannot be acquired with the help of a tutorial but should be trained during practical experience.

Teh Ru Qian

Although the video was only about 2.5 hours, it took me almost 3 days to complete the project due to several challenges along the way. I faced many issues such as account limitations during system setup, missing source files and many errors while running and debugging the pipeline. One of the problems occurred in the ForEach activity, where the copy_data tasks were designed to transfer data from the data lake. Initially, some of the copy_data activities succeeded while others failed, which made the issue difficult to diagnose. I spent lot of time debugging before realizing that the failures were caused by the pipeline running tasks concurrently. After modifying the settings to execute the tasks sequentially instead of in parallel, the pipeline finally ran successfully. These challenges became the most valuable part of the learning experience, as they strengthened my troubleshooting skills and gave me insight into solving real-world problems. Through this project, I understand how end-to-end data pipelines work, the role of Azure services and the important of clean, well-structured data in efficiently processing large datasets.

LING YU QIAN

The task of navigating this end-to-end pipeline posed unique architectural issues, especially security settings and identity management. In the process of setting up the cloud-native architecture, one of the biggest challenges was figuring out how to set up Role-Based Access Control (RBAC) and securely connect our services via Microsoft Entra ID and Azure Key Vault. In particular, to enable Azure Databricks and Synapse Analytics to read and write onto Data Lake smoothly, it was necessary to have specific Service Principal permissions. First, the access was denied as the definite role of the Storage Blob Data Contributor was not distributed and inherited throughout the resource group.This was solved with the help of an extensive search in the identity management documentation of Azure that allowed us to systematically debug our access policy and provide appropriate cross-service access without exposing credentials. This phase involved discuss question with Yu Han and Ru Qian; as we would have to methodically follow error logs jointly to have a clear understanding that pipeline breaks were caused either by compute failures or concealed authentication blocks.The structured debugging methodologies that I have used in my earlier rapid-prototyping environments and in my Year 3 engineering coursework were useful to isolate these faults in an efficient manner. The shift of the previously acquired experience of working with general cloud structures, to the peculiarities of the ecosystem of the Azure cloud, made my technical vision very broad. This project reinforced that a good data engineering project is as much about creating secure, well-managed infrastructure as it is about creating efficient transformation code.